

Item	Quantity	Unit Cost	Source	Phase	Status	Notes
Electronics						
Raspberry Pi 5	1	\$115.92	Amazon	1	Own	Main computer
RPi 5 Power Supply	1	\$14.46	Vilros	1	Own	27W USB-C, 5.1V, 5A
M.2 HAT+	1	\$23.99	Amazon	1	Own	Intermediary between RPi 5 and NVME for SSD
Active Cooler	1	\$11.89	Amazon	1	Own	Thermal management
Raspberry Pi NVMe SSD	1	\$89.99	Vilros	1	Own	Ubuntu boot drive
64 GB SSD	1	\$0.00	Office	1	Own	For practicing booting and Linux environment before NVMe SSD arrives
ELEGOO Sensor Kit	1	\$50.87	Amazon	1	Own	Miscellaneous sensors
RPi Camera Module 3 (Standard)	1	\$40.00	Vilros	1	Own	Camera with 12-megapixel Sony IMX708 image sensor module
Arduino Due	1	\$0.00	Office	1	Own	MCU
Arduino Mega Prototype Shield V3	1	\$0.00	Elegoo Kit	1	Own	For wiring and pass through of the motor controller
Adafruit 9-DOF Absolute Orientation IMU Fusion Breakout - BNO055 - STEMMA QT / Qwiic	1	\$29.95	Adafruit	1	Own	9-DOF sensor, MEMS accelerometer, magnetometer, and gyroscope, and putting them on a single die with a high-speed ARM Cortex-M0 based processor to digest all the sensor data
DROK DC Buck Converter, 5.3V-32V to 1.2V-32V 12A Adjustable Power Supply	1	\$17.49	Amazon	1	Own	Adjustable voltage buck converter for the RasPi 5. I needed it to be adjustable instead of set at 5V because it is reported the RasPi 5 can brown out when it gets marginally less than 5V.
Pololu Dual VNH5019 Motor Driver Shield for Arduino	1	\$39.95	Pololu	1	Ordered	This shield makes it easy to control two high-power DC motors with your Arduino or Arduino-compatible board. Its dual robust VNH5019 motor drivers operate from 5.5 to 24 V and can deliver a continuous 12 A (30 A peak) per motor, or a continuous 24 A (60 A peak) to a single motor connected to both channels. These great drivers also offer current-sense feedback and accept ultrasonic PWM frequencies for quieter operation. The Arduino pin mappings can all be customized if the defaults are not convenient, and the motor driver control lines are broken out along the left side of the shield for general-purpose use without an Arduino.
74LVC245 - Breadboard Friendly 8-bit Logic Level Shifter	2	\$1.50	Adafruit	1	Ordered	This chip solves the problem of how to interface 3.3V logic devices to a 5.0V logic chip such as the Arduino. Most 3.3V devices do not like being run with 5V signals and can be damaged or flaky. The 74LVC245 is designed so that even when it runs at 1.8V, it still happily accepts 5V signals in one pin and converts it to a lower logic level on the opposite pin. This chip is only for digital signals, it doesn't convert analog signals (you need an op-amp for that). It has 8 pipes it can convert but it won't work with bi-directional/pull-up based devices such as I2C or 1-Wire. It does work great for SPI, Serial, Parallel bus, and other logic interfaces.
IC Socket - for 20-pin 0.3" Chips - Pack of 3	1	\$0.95	Adafruit	1	Ordered	For connecting the Logic Level Shifter to the Due for Encloder communication
TBD RPLidar A1M8	1	\$0.00		2		
TBD 18650 Battery Pack (3S)	1	\$0.00		3		
TBD Battery Monitor	1	\$0.00		3		

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TBD Emergency Stop Button	1	\$0.00		3		
Arduino Mega 2560	4	\$0.00	Elegoo-Kit	4	Own	Motor-controller
Mechanical Components						
50:1 Metal Gearmotor 37Dx70L mm 12V with 64 CPR Encoder (Helical Pinion)	2	\$70.95	Pololu	1	Own	12V brushed DC motor with 16mm long, 6mm diameter D-shaped shaft, and encoder
Pololu Wheel 80×10mm Pair - Black	1	\$8.75	Pololu	1	Own	silicone tires and measure 80 mm (3.15") in diameter, and they press-fit onto 3mm D shafts
Pololu Universal Aluminum Mounting Hub for 6mm Shaft, #4-40 Holes (2-Pack)	1	\$12.95	Pololu	1	Own	Universal aluminum mounting hubs allow you to mount custom wheels and mechanisms to 6 mm diameter motor shafts
Pololu Ball Caster with 3/4" Plastic Ball	1	\$5.49	Pololu	1	Own	Light ball caster uses a 3/4" diameter plastic ball. The height of the assembled kit ranges from 0.9" to 1.2", depending on the combination of included spacers
Mini Panel Mount SPDT Toggle Switch	1	\$0.95	Adafruit	1	Own	Toggle switch that can power on and off the robot. Up to 1A at 24V recommended.
TBD Chassis	1	\$0.00	3D Printed	1		
Wiring and Connectors						
JST SH-Style Cable, 5-Pin, Female-Female, 16cm	3	\$3.06	Pololu	1	Own	This 5-conductor, 16 cm (6.3") cable has 28 AWG wires and 5-pin female JST SH-type connectors (1mm pitch) on both ends
JST SH-Style Cable, 5-Pin, Single-Ended Female, 12cm	3	\$1.77	Pololu	1	Own	This 5-conductor, 12 cm (4.5") cable has 28 AWG wires and a 5-pin female JST SH-type connector (1mm pitch) on one end. The other ends of the wires are unterminated; they can be cut to length to match your application.
2x20 Extra Tall Female 0.1" Pitch Stacking Header	8	\$9.79	Amazon	1	Own	23mm tall header that actually passes through the M.2 HAT+
STEMMA QT / Qwiic JST SH 4-pin Cable - 100mm Long	3	\$0.95	Adafruit	1	Own	fitted with JST-SH female 4-pin connectors on both ends
STEMMA QT / Qwiic JST SH 4-Pin Cable - 200mm Long	2	\$1.25	Adafruit	1	Own	fitted with JST-SH female 4-pin connectors on both ends
16AWG Silicone Wire 10ft	1	\$5.55	Amazon	1	Own	Will use to connect the battery to the motors and the buck converter
ALITOVE 3 Wire Splice Connectors 10pcs	1	\$7.99	Amazon	1	Own	For quick prototyping wire harnesses (avoids soldering and crimping)
Weewooday 10 Pieces 12-Position Dual Row White Screw Terminal Strip	1	\$6.99	Amazon	1	Own	For quick prototyping wire harnesses (avoids soldering and crimping)

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Waterproof Medium Inline Fuse Holder	1	\$4.89	Amazon	1	Own	To place near the battery for downstream electrical protection
80Pcs Standard Fuse Kit	1	\$4.99	Amazon	1	Own	Will use the 10A fuse with the inline fuse holder
Equipment						
FNIRSI® HS-02 Intelligent Soldering Iron	1	\$99.99	Fnirsi	1	Own	HS-02A is equipped with 6 soldering iron tips (K, KU, JS, I, C2, B). HS-02B is equipped with 3 soldering iron tips (I, IS, K). The integrated design allows the soldering iron to heat up quickly and maintain a stable temperature. And the 100 W charging adapter allows you to use it immediately.
Solder 0.032"	1	\$5.45	Amazon	1	Own	RA Flux Core
Soldering Iron Tip Cleaner	1	\$5.99	Amazon	1	Own	
Solder Wick & Solder Wire & Solder Sucker	1	\$8.99	Amazon	1	Own	
NICE-POWER DC Power Supply Variable 30V 10A	1	\$39.87	Amazon	1	Own	Adjustable Switching Regulated DC Bench Power Supply with Output Switch High Precision 4-Digit LED Display, Benchtop Lab Power Supplies with 5V/2A USB Port
Fun Unecessary Pick Ups						
Insulated Silicone Rework Mat - 34cm x 23cm Work Surface - Blue	1	\$9.95	Adafruit	0	Own	Non-slip, ESD safe surface to prevent damaging boards accidentally on my plastic desk surface (in the middle of a dry NYC winter)
INA219 High Side DC Current Sensor Breakout - 26V ±3.2A Max - STEMMA QT	1	\$9.95	Adafruit	0	Own	Current measuring chip to use instead of blowing out multimeters
Adafruit APDS9960 Proximity, Light, RGB, and Gesture Sensor - STEMMA QT / Qwiic	1	\$7.50	Adafruit	0	Own	Basic gesture sensing, RGB color sensing, proximity sensing, or ambient light sensing
USB to TTL Serial Cable - Debug / Console Cable for Raspberry Pi	1	\$9.95	Adafruit	0	Own	Easiest way ever to connect to your microcontroller/Raspberry Pi/WiFi router serial console port
GPIO Header for RaspberryPi A+/B+/Pi 2/Pi 3 - Tall 2x20 Female Header	1	\$2.95	Adafruit	1	Own	TOO SHORT This spacer will extend the headers through the M.2 HAT+ on my RasPi 5 so I can actually use them
GPIO Stacking Header for Pi A+/B+/Pi 2/Pi 3 - Extra-long 2x20 Pins	1	\$2.75	Adafruit	1	Own	TOO SHORT This spacer will extend the headers through the M.2 HAT+ on my RasPi 5 so I can actually use them
Pololu G2 High-Power Motor Driver 18v17	1	\$44.95	Pololu	1	Own	Dual motor driver, 6.5V-30V, 17A